



# **GCSE MARKING SCHEME**

**SUMMER 2024**

**GCSE  
BIOLOGY – UNIT 1  
3400U10-1 AND 3400UA0-1**

## About this marking scheme

The purpose of this marking scheme is to provide teachers, learners, and other interested parties, with an understanding of the assessment criteria used to assess this specific assessment.

This marking scheme reflects the criteria by which this assessment was marked in a live series and was finalised following detailed discussion at an examiners' conference. A team of qualified examiners were trained specifically in the application of this marking scheme. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners. It may not be possible, or appropriate, to capture every variation that a candidate may present in their responses within this marking scheme. However, during the training conference, examiners were guided in using their professional judgement to credit alternative valid responses as instructed by the document, and through reviewing exemplar responses.

Without the benefit of participation in the examiners' conference, teachers, learners and other users, may have different views on certain matters of detail or interpretation. Therefore, it is strongly recommended that this marking scheme is used alongside other guidance, such as published exemplar materials or Guidance for Teaching. This marking scheme is final and will not be changed, unless in the event that a clear error is identified, as it reflects the criteria used to assess candidate responses during the live series.

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**WJEC GCSE BIOLOGY**  
**SUMMER 2024 MARK SCHEME**

Recording of marks

Examiners must mark in red ink.

One tick must equate to one mark (apart from the questions where a level of response mark scheme is applied). Question totals should be written in the box at the end of the question.

Question totals should be entered onto the grid on the front cover and these should be added to give the script total for each candidate.

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer. Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statements.

### Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

|     |   |                       |
|-----|---|-----------------------|
| cao | = | correct answer only   |
| ecf | = | error carried forward |
| bod | = | benefit of doubt      |

# FOUNDATION TIER

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                | Marks Available |     |     |       |       |      |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 1        | (a) |  |  | A right atrium labelled (1)<br>B left ventricle labelled (1)                                                                                                                                                                                                                                                   | 2               |     |     | 2     |       |      |
|          | (b) |  |  | <div> <div> Pulmonary vein<br/>Aorta<br/>Vena cava<br/>Pulmonary artery </div> <div> From the body to the heart<br/>From the heart to the body<br/>From the heart to the lungs<br/>From the lungs to the heart </div> </div> <p>3/4 correct for 3 marks<br/>2 correct for 2 marks<br/>1 correct for 1 mark</p> | 3               |     |     | 3     |       |      |
|          | (c) |  |  | Prevent the backflow of blood underlined                                                                                                                                                                                                                                                                       | 1               |     |     | 1     |       |      |
|          | (d) |  |  | Muscle (1)<br>Pump (1)<br>Coronary (1)                                                                                                                                                                                                                                                                         | 3               |     |     | 3     |       |      |
|          |     |  |  | Question 1 total                                                                                                                                                                                                                                                                                               | 9               | 0   | 0   | 9     | 0     | 0    |

| Question       |                  |                 |     | Marking details                                                                                                                                                                                                                            | Marks Available |                  |                 |          |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|----------------|------------------|-----------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-----------------|----------|-------|------|--------|----|----|----------------|------|---|---|--|--|---|--|--|
|                |                  |                 |     |                                                                                                                                                                                                                                            | AO1             | AO2              | AO3             | Total    | Maths | Prac |        |    |    |                |      |   |   |  |  |   |  |  |
| 2              | (a)              |                 |     | X bronchiole<br>Y alveolus                                                                                                                                                                                                                 | 2               |                  |                 | 2        |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                | (b)              | (i)             | I   | Oxygen                                                                                                                                                                                                                                     | 1               |                  |                 | 1        |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                |                  |                 | II  | Diffusion                                                                                                                                                                                                                                  | 1               |                  |                 | 1        |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                |                  |                 | III | Respiration                                                                                                                                                                                                                                | 1               |                  |                 | 1        |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                |                  | (ii)            |     | <b>Any two (x1) from:</b><br>{Thin / one cell thick} alveolus walls (1)<br>{Thin / one cell thick} capillary walls (1)<br>Capillary very close to alveolus / good blood supply (1)<br>Large surface area (1)<br>Moist (1)                  |                 | 2                |                 | 2        |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                | (c)              |                 |     | <table><tr><th>Gas</th><th>Inspired air (%)</th><th>Expired air (%)</th></tr><tr><td>Nitrogen</td><td>78</td><td>78</td></tr><tr><td>Oxygen</td><td>21</td><td>16</td></tr><tr><td>Carbon dioxide</td><td>0.04</td><td>4</td></tr></table> | Gas             | Inspired air (%) | Expired air (%) | Nitrogen | 78    | 78   | Oxygen | 21 | 16 | Carbon dioxide | 0.04 | 4 | 3 |  |  | 3 |  |  |
| Gas            | Inspired air (%) | Expired air (%) |     |                                                                                                                                                                                                                                            |                 |                  |                 |          |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
| Nitrogen       | 78               | 78              |     |                                                                                                                                                                                                                                            |                 |                  |                 |          |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
| Oxygen         | 21               | 16              |     |                                                                                                                                                                                                                                            |                 |                  |                 |          |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
| Carbon dioxide | 0.04             | 4               |     |                                                                                                                                                                                                                                            |                 |                  |                 |          |       |      |        |    |    |                |      |   |   |  |  |   |  |  |
|                |                  |                 |     | Question 2 total                                                                                                                                                                                                                           | 8               | 2                | 0               | 10       | 0     | 0    |        |    |    |                |      |   |   |  |  |   |  |  |

| Question         |     |       |  | Marking details                                                                                                                                                                                                                                                              | Marks Available |     |     |       |       |      |
|------------------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|                  |     |       |  |                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3                | (a) | (i)   |  | 12.3 (2)<br><b>Award 1 mark for</b><br>49/4<br>12.25                                                                                                                                                                                                                         |                 | 2   |     | 2     | 2     |      |
|                  |     | (ii)  |  | (Used for) energy                                                                                                                                                                                                                                                            | 1               |     |     | 1     |       |      |
|                  | (b) | (i)   |  | 17 g                                                                                                                                                                                                                                                                         |                 | 1   |     | 1     | 1     |      |
|                  |     | (ii)  |  | <b>Any two (x1) from:</b><br>Obesity (1)<br>Heart disease (1)<br>Circulatory disease / CVD (1)<br>Diabetes (1)                                                                                                                                                               | 2               |     |     | 2     |       |      |
|                  | (c) | (i)   |  | <b>Any two (x1) from:</b><br>(wear) eye protection / goggles / safety glasses (1)<br>care with mounted needle (1)<br>hold mounted needle at arm's length / keep hands away from flame (1)<br>hair tied back (1)<br>stand up (1)<br>Allow apparatus to cool before moving (1) |                 | 2   |     | 2     |       | 2    |
|                  |     | (ii)  |  | Thermometer                                                                                                                                                                                                                                                                  |                 | 1   |     | 1     |       | 1    |
|                  | (d) | (i)   |  | 21 (°C)                                                                                                                                                                                                                                                                      |                 | 1   |     | 1     | 1     | 1    |
|                  |     | (ii)  |  | 3528 = 2 marks<br><b>If incorrect award 1 mark for</b><br><u>20 x 21 x 4.2</u><br>0.5<br>Allow ecf                                                                                                                                                                           |                 | 2   |     | 2     | 2     | 2    |
|                  |     | (iii) |  | {Heat / energy} lost (to the surroundings) /<br>the crisp was not all burnt                                                                                                                                                                                                  |                 |     | 1   | 1     |       | 1    |
| Question 3 total |     |       |  |                                                                                                                                                                                                                                                                              | 3               | 9   | 1   | 13    | 6     | 6    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                            | Marks Available |          |          |          |          |          |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) | (i)  |  | False<br>True<br>True<br>False<br>All correct for 3 marks<br>3 correct for 2 marks<br>2 correct for 1 marks<br>0/1 correct = 0 marks                                                                                                                                                                                                                                                                                       |                 | 3        |          | 3        |          |          |
|          |     | (ii) |  | {Mercury / it} is {released / dumped} into {water / sea / ocean}<br>(by factory) (1)<br><u>Algae</u> took up the mercury (1)                                                                                                                                                                                                                                                                                               |                 | 1        | 1        | 2        |          |          |
|          | (b) | (i)  |  | 12 = 2 marks<br><br>1200 / 100 = 1 mark                                                                                                                                                                                                                                                                                                                                                                                    |                 | 2        |          | 2        | 2        |          |
|          |     | (ii) |  | <b>Any two (x1) from:</b> <ul style="list-style-type: none"> <li>Mercury {accumulates in living tissue / cannot be broken down} / bioaccumulation (1)</li> <li>Mercury (is taken up by algae in the sea) and then passed along the food chain / concentration of mercury increases along the food chain (1)</li> <li>Until it reaches toxic levels / (cats get) lethal dose/ level becomes too concentrated (1)</li> </ul> |                 |          | 2        | 2        |          |          |
|          |     |      |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                    | <b>0</b>        | <b>6</b> | <b>3</b> | <b>9</b> | <b>2</b> | <b>0</b> |



| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks Available |     |     |       |       |      |
|----------|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        |  |  |  | <p><b>Indicative content:</b></p> <p>A. Carbon transferred from air to plants / carbon dioxide taken in by plants</p> <p>B. {by / for} photosynthesis / <b>W</b> is photosynthesis</p> <p>C. Carbon transferred from plants to animals.</p> <p>D. by feeding / <b>Y</b> is feeding.</p> <p>E. Carbon transferred from {animals / plants / decomposers} to the air / carbon dioxide is released by {animals / plants / decomposers}</p> <p>F. by respiration / <b>X</b> is respiration.</p> <p>G. Carbon {transferred into air / released} by combustion / <b>Z</b> is combustion.</p> <p>H. (The names of the decomposers are) bacteria.</p> <p>I. and fungi</p> <p><b>5-6 marks</b><br/>At least seven points from the indicative content<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>At least four points from the indicative content<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> | 2               | 4   |     | 6     |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |          |          |          |          |          |
|----------|--|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          |  |  |  | <b>1-2 marks</b><br>At least one point from the indicative content<br><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i><br><br><b>0 marks</b><br><i>No attempt made or no response worthy of credit.</i> |                 |          |          |          |          |          |
|          |  |  |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                                                                                                                       | <b>2</b>        | <b>4</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                      | Marks Available |     |     |       |       |      |
|----------|-----|-------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6        | (a) |       |  | Proteins (1)<br>Fats / lipids / oils (1)                                                                                                                                                                                                                                                                             | 2               |     |     | 2     |       |      |
|          | (b) | (i)   |  | <ul style="list-style-type: none"><li>40°C (1)</li><li>Because it removed the most stain /<br/>The stain was the {lightest / most faded / least visible} /<br/>Most egg removed /<br/>Fabric was the cleanest /<br/>owtte (1)</li></ul>                                                                              |                 |     | 2   | 2     |       | 2    |
|          |     | (ii)  |  | <b>Any three from (x1)</b> <ul style="list-style-type: none"><li>Stain not removed (1)</li><li>Too hot / temperature too high / above optimum<br/>temperature (1)</li><li>{Enzymes / active site} are <u>denatured</u> (1)</li><li>Active site changes <u>shape</u> (1)</li><li>Substrate does not fit (1)</li></ul> |                 |     | 3   | 3     |       |      |
|          | (c) | (i)   |  | As temperature increased {the more stain was removed / less<br>egg remained}                                                                                                                                                                                                                                         |                 | 1   |     | 1     |       | 1    |
|          |     | (ii)  |  | as a control / to compare results / owtte                                                                                                                                                                                                                                                                            |                 | 1   |     | 1     |       | 1    |
|          |     | (iii) |  | can wash at lower temperatures / can wash at 40°C instead of<br>60 °C (1)<br>save <u>energy</u> / ref energy sources e.g. less electricity (1)                                                                                                                                                                       |                 | 1   | 1   | 2     |       |      |
|          | (d) |       |  | <b>Any two (x1) from</b><br>{Area / size} of {egg / staining} (1)<br>{Mass / concentration} of washing powder (1)<br>volume of water in beaker (1)<br>type of fabric (1)<br>pH (1)                                                                                                                                   |                 |     | 2   | 2     |       | 2    |
|          |     |       |  | Question 6 total                                                                                                                                                                                                                                                                                                     | 2               | 3   | 8   | 13    | 0     | 6    |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                       | Marks |     |     |       |       |      |
|----------|-----|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                       | AO1   | AO2 | AO3 | Total | Maths | Prac |
| 7/1      | (a) | (i)   |  | grass                                                                                                                                                                                                                                                                                                 | 1     |     |     | 1     |       |      |
|          |     | (ii)  |  | Producer<br>Secondary consumer<br>Carnivore<br><u>Herbivore</u><br><u>Primary consumer</u><br><br><b>Both correct for 1 mark</b>                                                                                                                                                                      |       | 1   |     | 1     |       |      |
|          |     | (iii) |  | grass → zebra → lion → fleas<br><br><b>1 mark</b> for organisms in correct order joined by arrows                                                                                                                                                                                                     |       | 1   |     | 1     |       |      |
|          |     | (iv)  |  | Sun / light / sunlight                                                                                                                                                                                                                                                                                | 1     |     |     | 1     |       |      |
|          |     | (v)   |  | <b>Any one (x 1) from:</b><br>(energy is) used in {growth / repair} (1)<br>(energy is) lost in {waste materials / named waste materials / respiration / movement / heat} (1)<br>not all of the organism is {consumed / digested} (1)                                                                  | 1     |     |     | 1     |       |      |
|          |     | (vi)  |  | <b>Any one (x 1) from:</b><br>because {pyramids of numbers / it} does not take into account the {size / mass} of the organisms (1)<br>fleas are smaller than lions (1)<br>there are more fleas than lions (1)<br>lots of fleas can live on one lion (1)<br>more tertiary than secondary consumers (1) |       |     | 1   | 1     |       |      |

| Question |  |       |  | Marking details                                                                                                                                                                                                             | Marks    |          |          |          |          |          |
|----------|--|-------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|
|          |  |       |  |                                                                                                                                                                                                                             | AO1      | AO2      | AO3      | Total    | Maths    | Prac     |
|          |  | (vii) |  | <ul style="list-style-type: none"> <li>pyramid shaped with 4 bars (each bar smaller than the one below). (1)</li> <li>Labelled with the organisms names (1)</li> </ul> <p>2<sup>nd</sup> MP linked to 1<sup>st</sup> MP</p> |          | 2        |          | 2        |          |          |
|          |  |       |  | <b>Total for question 7/1</b>                                                                                                                                                                                               | <b>3</b> | <b>4</b> | <b>1</b> | <b>8</b> | <b>0</b> | <b>0</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                | Marks Available |          |          |           |          |          |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 8/2      | (a) |       |  | Nitrates / N (1)<br>Phosphates / P (1)<br>Potassium / K (1)                                                                                                                                                                                                                                                                                                                                                                    | 3               |          |          | 3         |          |          |
|          | (b) |       |  | IV – (addition of) fertiliser / treatment of tomato plants (1)<br>DV – mass of tomatoes (1)                                                                                                                                                                                                                                                                                                                                    | 2               |          |          | 2         |          | 2        |
|          | (c) | (i)   |  | 46                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |          | 1        | 1         | 1        |          |
|          |     | (ii)  |  | 30 = 2 marks<br>29.6 = 1 mark                                                                                                                                                                                                                                                                                                                                                                                                  |                 | 2        |          | 2         | 2        | 2        |
|          |     | (iii) |  | Adding fertiliser:<br>increased the mass /<br>made {the tomatoes / them} {bigger / grow more}<br>Accept reverse argument for no fertiliser<br>Ignore more tomatoes                                                                                                                                                                                                                                                             |                 |          | 1        | 1         |          | 1        |
|          | (d) | (i)   |  | <b>Any one (x1) from:</b><br>{species / type} of tomato (plant) (1)<br>volume of water (1)<br>light (1)<br>soil (1)<br>temperature (1)<br>time that the plants grew for / plants grew for 3 months (1)                                                                                                                                                                                                                         |                 | 1        |          | 1         |          | 1        |
|          |     | (ii)  |  | Reason for inaccuracy <ul style="list-style-type: none"> <li>small sample size / only 10 /</li> <li>his sample may not have been random/<br/>there could have been larger tomatoes on the plants without fertiliser that he did not pick / owtte</li> </ul> Suggestion for improvement <ul style="list-style-type: none"> <li>{measure / use} (the mass of) {all / more} of the tomatoes /<br/>increase sample size</li> </ul> |                 | 1        | 1        | 2         |          | 2        |
|          |     |       |  | <b>Question 8/2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                      | <b>5</b>        | <b>4</b> | <b>3</b> | <b>12</b> | <b>3</b> | <b>9</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) |      |  | Red blood cell / RBC correctly labelled<br>White blood cell / WBC correctly labelled<br><b>Both correct for 1 mark</b><br>Ignore direction of arrow.<br>Arrow must touch cells                                                                               | 1               |     |     | 1     |       |      |
|          | (b) | (i)  |  | <b>3 marks</b><br>57<br><br><b>2 marks for sight of</b><br>57.0877944 / 57.33333333 (if rounded earlier)<br>$\frac{2666}{4670} \times 100$<br>$\frac{43}{75.3225806} \times 100$<br><br><b>1mark for sight of</b><br>43 x 62 = 2 666<br>4670/62 = 75.3225806 |                 | 3   |     | 3     | 3     |      |
|          |     | (ii) |  | {Transport / carries} {CO <sub>2</sub> / soluble food / urea / hormones / antibodies / glucose / amino acids / vitamins / minerals / fats/ products of digestion} / distribution of heat (1)<br>Ignore: carry {blood cells / nutrients}                      | 1               |     |     | 1     |       |      |

|  |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |          |          |          |          |          |
|--|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|----------|----------|----------|
|  | (c) | (i)  | <p><b>Any two (x1) from:</b></p> <ul style="list-style-type: none"> <li>least red blood cells /<br/>RBC below normal range /<br/>RBC lower than other patients /<br/>less RBC than other patients /<br/>use of data, e.g. RBC is 4.4 and the range is 4.5 - 5.2 (1)</li> <li>Least haemoglobin /<br/>haemoglobin below normal range /<br/>haemoglobin lower than other patients /<br/>less haemoglobin than other patients /<br/>use of data, e.g. haemoglobin is 10.9 and the range is 13.2 - 14.8 (1)</li> <li>cannot {carry / transport} enough oxygen /<br/>{red blood cells / haemoglobin} {carry / transport} oxygen (1)</li> </ul> | 1        | 1        |          | 2        |          |          |
|  |     | (ii) | <ul style="list-style-type: none"> <li>{least / lowest concentration of / lowest number of} white blood cells /<br/>less white blood cells than the other patients (must be comparative) (1)</li> <li>{defend / protects} against {disease / pathogens / bacteria / fungi / viruses / microbes / microorganisms} /<br/><br/>{Kill / destroy / engulf} {bacteria / fungi / viruses / microbes / microorganisms / pathogens} / correct ref to antibody production(1)<br/>Ignore references to infection</li> </ul>                                                                                                                          | 1        | 1        |          | 2        |          |          |
|  |     |      | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b> | <b>5</b> | <b>0</b> | <b>9</b> | <b>3</b> | <b>0</b> |



| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                       | Marks available |          |          |          |          |          |
|----------|-----|--|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 4        | (a) |  |  | Indicator species (1)                                                                                                                                                                                                                                                                                 | 1               |          |          | 1        |          |          |
|          | (b) |  |  | Location A = <u>high</u> (1)<br>Location B = <u>low</u> (1)                                                                                                                                                                                                                                           |                 |          | 2        | 2        |          | 2        |
|          | (c) |  |  | <ul style="list-style-type: none"> <li>• (Sun)light is blocked / less light reaches the plant (1)</li> <li>• (aquatic) plants die (1)<br/>Ignore algae</li> <li>• {decomposers / bacteria} {decompose / decay} (dead material) (1)</li> <li>• (decomposers) use oxygen for respiration (1)</li> </ul> | 4               |          |          | 4        |          |          |
|          |     |  |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                               | <b>5</b>        | <b>0</b> | <b>2</b> | <b>7</b> | <b>0</b> | <b>2</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |     |     |       |       |      |
|----------|-----|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 5        | (a) | (i)   |  | A = guard (cell) (1)<br>B = epidermal (cell) / epidermis (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 | 2   |     | 2     |       |      |
|          |     | (ii)  |  | 24mm(1)<br>Accept 23 - 25mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1               |     |     | 1     |       | 1    |
|          |     | (iii) |  | <b>For 24mm</b><br>Correct answer = 1200 (2)<br><b>If incorrect award 1 mark for sight of</b><br><b>Correct conversion</b><br>24mm to 24000µm / 20µm to 0.02mm (must show evidence of units)<br><b>Correct substitution into a formula</b><br>24000/20 OR 24/0.02<br><br><b>For 25mm</b><br>Correct answer = 1250 (2)<br><b>If incorrect award 1 mark for sight of</b><br><b>Correct conversion</b><br>25mm to 25000µm / 20µm to 0.02mm (must show evidence of units)<br><b>Correct substitution into a formula</b><br>25000/20 OR 25/0.02 |                 | 2   |     | 2     | 2     | 2    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                   | Marks available |          |          |          |          |          |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                   | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          |     |      |  | <b>For 23mm</b><br>Correct answer = 1150 (2)<br><b>If incorrect award 1 mark for sight of</b><br><b>Correct conversion</b><br>23mm to 23000µm / 20µm to 0.02mm (must show evidence of units)<br><b>Correct substitution into a formula</b><br>23000/20 OR 23/0.02<br><br>Accept ecf from (a) (ii) |                 |          |          |          |          |          |
|          | (b) | (i)  |  | Stomata open {in response to / during} {light / day} / ORA(1)<br>Ignore reference to stomatal size                                                                                                                                                                                                |                 |          | 1        | 1        |          |          |
|          |     | (ii) |  | (stomata) {control / regulate / reduce} {water loss / transpiration}<br>(1)<br><br>Reject prevent water loss                                                                                                                                                                                      |                 | 1        |          | 1        |          |          |
|          |     |      |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                           | <b>1</b>        | <b>5</b> | <b>1</b> | <b>7</b> | <b>2</b> | <b>3</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                               | Marks available |          |          |          |          |          |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 6        | (a) |      |  | <b>Aerobic</b><br>Water / H <sub>2</sub> O / H <sub>2</sub> O / H <sub>2</sub> O (1)<br>Carbon dioxide / CO <sub>2</sub> / CO <sub>2</sub> / CO <sub>2</sub> (1)<br><br><b>Anaerobic</b><br>Lactic acid / lactate (1)                                                                         | 3               |          |          | 3        |          |          |
|          | (b) |      |  | <b>Any one (x1) from</b> <ul style="list-style-type: none"> <li>{lactic acid / lactate} causes {pain / (muscle) fatigue / cramps} (1)</li> <li>Oxygen debt (1)</li> <li>provides less {energy / ATP} / less efficient than aerobic respiration (1)</li> </ul>                                 | 1               |          |          | 1        |          |          |
|          | (c) | (i)  |  | <b>Any one (x1) from</b> <ul style="list-style-type: none"> <li>takes time for respiration to {produce ATP / release energy} (1)</li> <li>provide {energy / ATP} before respiration starts (1)</li> <li>There is no {ATP / energy} from respiration at the start (of exercise) (1)</li> </ul> |                 |          | 1        | 1        |          |          |
|          |     | (ii) |  | Aerobic respiration produces {more / most} ATP / aerobic respiration produces 38 ATP compared to anaerobic which produces 2 ATP / owtte / ORA (1)<br><br>Glucose (molecule) not fully broken down (in anaerobic respiration) / ORA (1)                                                        |                 | 1        | 1        | 2        |          |          |
|          |     |      |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                       | <b>4</b>        | <b>1</b> | <b>2</b> | <b>7</b> | <b>0</b> | <b>0</b> |

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |      |  | <ul style="list-style-type: none"> <li>It is always higher in females (than males) / ORA (1)</li> <li>Both increase (and level off) (1)</li> <li>The difference between males and females has decreased / It has increased more for men (than women) (1)</li> </ul> Ignore {women / men} live longer / references to life expectancy                                                                       |                 | 3   |     | 3     |       |      |
|          | (b) |      |  | <b>Any two (x1) from:</b><br>Excess energy stored as fat (1)<br>Leading to obesity (1)<br>(Which could lead to)<br>heart disease / CVD / coronary heart disease / diabetes / any correct health related issue (1)                                                                                                                                                                                          | 2               |     |     | 2     |       |      |
|          | (c) | (i)  |  | <b>Any two (x1) from:</b> <ul style="list-style-type: none"> <li>{more / increased} {fruit / vegetables} (1)</li> <li>{more / increased} {fibre / vitamins / minerals} in diet (1)</li> <li>Contribute to a balanced diet / Prevent constipation / reduce {risk of / chance of} {bowel cancer / heart disease / high blood pressure / stroke} (1)</li> </ul> Accept ref to preventing dietary deficiencies |                 | 2   |     | 2     |       |      |
|          |     | (ii) |  | <b>Any two (x1) from:</b><br>(less) (type 2) diabetes (1)<br>(less) obesity (1)<br>(less) tooth decay (1)                                                                                                                                                                                                                                                                                                  | 2               |     |     | 2     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                | Marks available |          |          |           |          |          |
|----------|-----|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |  |                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (d) |  |  | <p>Strong evidence because:<br/>           Large number of researchers / 7 000 researchers (1)<br/>           in {a large number of countries / 156 countries} (1)</p> <p>Max one mark if do not refer to strength of evidence / if judgement is incorrect</p> |                 |          | 2        | 2         |          | 2        |
|          |     |  |  | <b>Question 7 total</b>                                                                                                                                                                                                                                        | <b>4</b>        | <b>5</b> | <b>2</b> | <b>11</b> | <b>0</b> | <b>2</b> |

| Question |     |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|-------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |  |                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8        | (a) | (i)   |  | Lock and key (1)                                                                                                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       |      |
|          |     | (ii)  |  | {Sequence / order / chain} of amino acids (in protein) / folding of {amino acid chain / protein} (1)                                                                                                                                                                                                                                                                                                          | 1               |     |     | 1     |       |      |
|          | (b) | (i)   |  | <ul style="list-style-type: none"> <li>Less (intensity of) light passing through (beaker / solution) / light {has been blocked / cannot pass through} / harder for light to pass through (1)</li> <li>Iodine has reacted with starch / solution in beaker has turned {blue / black / opaque / darker} (1)</li> </ul>                                                                                          |                 | 1   | 1   | 2     |       | 1    |
|          |     | (ii)  |  | 92 = 3 marks<br><b>If incorrect award 2 marks</b><br>Correct readings from graph <b>and</b> substitution into formula (620-160)/5 or 460/5<br><b>If incorrect award 1 mark</b><br>Correct readings from graph<br>620 <b>and</b> 160<br>460                                                                                                                                                                    |                 | 3   |     | 3     | 3     |      |
|          |     | (iii) |  | <ul style="list-style-type: none"> <li>carbohydrase (has been added / breaks down) (1)</li> <li>(so) starch (is broken down) to glucose (1)<br/>Accept maltose</li> <li>{More / easier for} light passing through (solution beaker) / solution becomes {lighter / less opaque} / solution turns from {blue / black} to {orange / brown} / {blue / black} colour {starts to clear / disappears} (1)</li> </ul> |                 | 2   | 1   | 3     |       | 3    |

| Question |  |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |           |          |          |
|----------|--|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (iv) |  | <b>Any two (x1) from:</b> <ul style="list-style-type: none"> <li>• {volume / 100cm<sup>3</sup>} of starch (1)</li> <li>• {volume / 20cm<sup>3</sup>} of carbohydrase (1)</li> <li>• {number / amount} of drops of iodine / volume of iodine (1)</li> <li>• {concentration / (1)%} of starch (1)</li> <li>• {concentration / (10)%} of carbohydrase (1)</li> <li>• concentration of iodine (1)</li> <li>• temperature (1)</li> <li>• pH (1)</li> <li>• distance of lamp from <u>sensor</u> / light intensity <u>of lamp</u> (1)</li> </ul> | 2               |          |          | 2         |          | 2        |
|          |  | (v)  |  | Line on graph remains at same level from 10 minutes (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |          | 1        | 1         |          |          |
|          |  |      |  | <b>Question 8 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>4</b>        | <b>6</b> | <b>3</b> | <b>13</b> | <b>3</b> | <b>6</b> |



| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
|----------|--|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9        |  |  |  | <p><b>Indicative content:</b></p> <p><i>Potato A</i></p> <p>A. {level / volume} of solution (in cavity) increases</p> <p>B. Lower water concentration in the potato compared to the {Petri dish / distilled water} / ORA.</p> <p>C. (so) water moves into the potato (from the Petri dish / distilled water)</p> <p>D1 Lower water concentration in the {cavity / sugar solution} compared to the potato / ORA</p> <p>OR</p> <p>D2 lower water concentration in {cavity / sugar solution} compared to {distilled water / Petri dish} / ORA</p> <p>E. (So) water moves into the {cavity / sugar solution}</p> <p>F. Correct use of the word osmosis /<br/>water has moved down a water {concentration / potential} gradient /<br/>water moves from a high concentration of water to a low concentration of water</p> <p>G. Any correct reference to {semi / partially / selectively} permeable membrane.</p> <p><i>Potato B</i></p> <p>H. This is a control (experiment)<br/>Reject reference to controlled</p> <p>I. to show that it is the presence of sugar solution (in the cavity of the fresh potato) that causes the effect observed / to show that the effect is caused by the independent variable</p> | 2               | 2   | 2   | 6     |       | 6    |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
|----------|--|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>5-6 marks</b><br/>At least seven points from the indicative content<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>At least four points from the indicative content<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> <p><b>1-2 marks</b><br/>At least one point from the indicative content<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |     |     |       |       |      |
|          |  |  |  | Question 9 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2               | 2   | 2   | 6     |       | 6    |

**FOUNDATION TIER****SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES**

| <b>Question</b> | <b>AO1</b> | <b>AO2</b> | <b>AO3</b> | <b>TOTAL MARK</b> | <b>MATHS</b> | <b>PRAC</b> |
|-----------------|------------|------------|------------|-------------------|--------------|-------------|
| <b>1</b>        | <b>9</b>   | <b>0</b>   | <b>0</b>   | <b>9</b>          | <b>0</b>     | <b>0</b>    |
| <b>2</b>        | <b>8</b>   | <b>2</b>   | <b>0</b>   | <b>10</b>         | <b>0</b>     | <b>0</b>    |
| <b>3</b>        | <b>3</b>   | <b>9</b>   | <b>1</b>   | <b>13</b>         | <b>6</b>     | <b>6</b>    |
| <b>4</b>        | <b>0</b>   | <b>6</b>   | <b>3</b>   | <b>9</b>          | <b>2</b>     | <b>0</b>    |
| <b>5</b>        | <b>2</b>   | <b>4</b>   | <b>0</b>   | <b>6</b>          | <b>0</b>     | <b>0</b>    |
| <b>6</b>        | <b>2</b>   | <b>3</b>   | <b>8</b>   | <b>13</b>         | <b>0</b>     | <b>6</b>    |
| <b>7</b>        | <b>3</b>   | <b>4</b>   | <b>1</b>   | <b>8</b>          | <b>0</b>     | <b>0</b>    |
| <b>8</b>        | <b>5</b>   | <b>4</b>   | <b>3</b>   | <b>12</b>         | <b>3</b>     | <b>9</b>    |
| <b>TOTAL</b>    | <b>32</b>  | <b>32</b>  | <b>16</b>  | <b>80</b>         | <b>11</b>    | <b>21</b>   |

## HIGHER TIER

### SUMMARY OF MARKS ALLOCATED TO ASSESSMENT OBJECTIVES

| Question     | AO1       | AO2       | AO3       | TOTAL MARK | MATHS    | PRAC      |
|--------------|-----------|-----------|-----------|------------|----------|-----------|
| 1            | 3         | 4         | 1         | 8          | 0        | 0         |
| 2            | 5         | 4         | 3         | 12         | 3        | 9         |
| 3            | 4         | 5         | 0         | 9          | 3        | 0         |
| 4            | 5         | 0         | 2         | 7          | 0        | 2         |
| 5            | 1         | 5         | 1         | 7          | 2        | 3         |
| 6            | 4         | 1         | 2         | 7          | 0        | 0         |
| 7            | 4         | 5         | 2         | 11         | 0        | 2         |
| 8            | 4         | 6         | 3         | 13         | 3        | 6         |
| 9            | 2         | 2         | 2         | 6          | 0        | 6         |
| <b>TOTAL</b> | <b>24</b> | <b>24</b> | <b>12</b> | <b>60</b>  | <b>8</b> | <b>13</b> |